# Highly Accurate Molecular Properties using High-Order Relativistic Coupled Cluster Theory

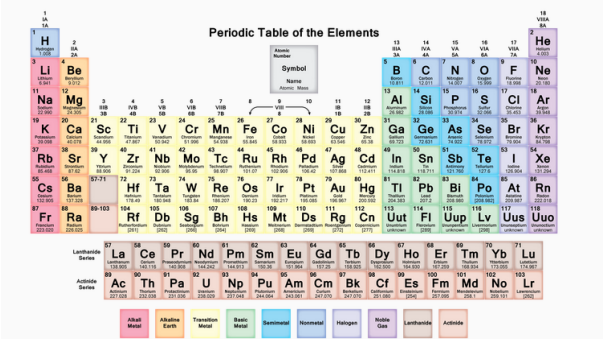
**Gabriele Fabbro**
Supervisor: Trond Saue
Laboratoire de Chimie et Physique Quantiques - UMR5626
Université de Toulouse

5 December 2025

Laboratoire de Chimie et Physique Quantiques

# Introduction

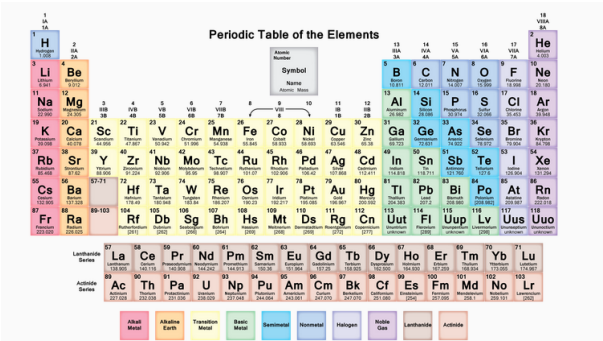
## Our playground




- Elements of the periodic table are described using quantum mechanics.

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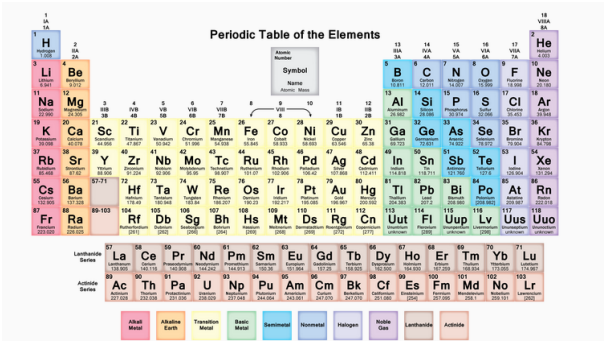
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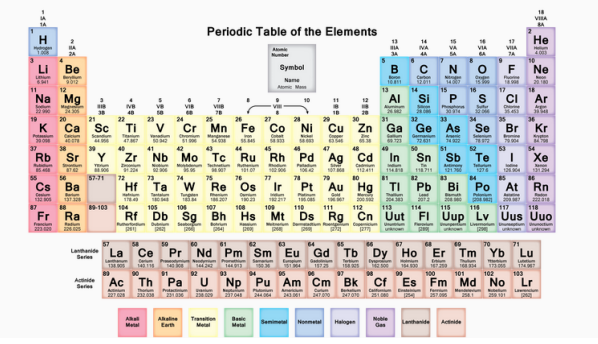
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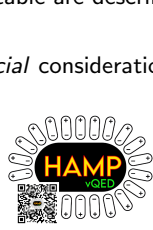

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  - **Correlation effects**
  - **Relativistic effects**
  - **QED effects**

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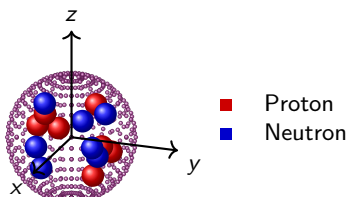



- A property that is particularly sensitive to these effects is the **electric field gradient (EFG)**.

# Introduction

## Nuclear quadrupole interaction (NQI)

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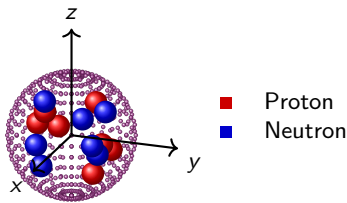

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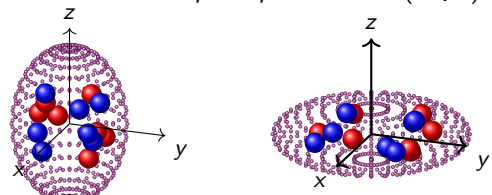
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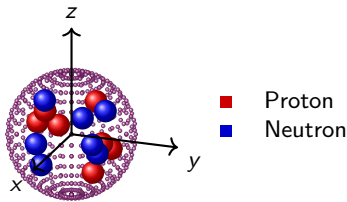
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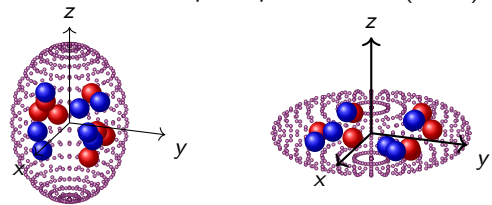
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**Nuclear quadrupole interaction (NQI)**

The NQM of the nucleus K couples with the EFG generated by the surrounding particles

$$\text{NQCC [in MHz]} = 234.9647 \times\ Q \text{ [in b]} \times\ q \text{ [in } E_h/a_0^2\text{]} \tag{1}$$

# Goals of the thesis

**Physical information**

By combining experiment and theory we may extract the NQM:

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This was pointed out by Dailey and Townes:

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What about heavy systems where we have *d* and *f* orbitals and relativistic effects?

### Development of high-order CC expectation values

### Beyond the Dailey & Townes model

# Beyond the Dailey-Townes Model: Chemical information from the Electric Field Gradient

G. Fabbro, J. Pototschnig and T. Saue, *J. Phys. Chem. A* **129**, 1006 (2025)

# Projection analysis

S. Dubillard, J.-B. Rota, T. Saue and K.Fægri, J. Chem. Phys. 124, 154307, (2006)

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$$|\psi_i^{MO}\rangle = \sum_A \sum_{p \in A} c_{p,i}^A |\varphi_p^A\rangle + |\varphi_i^{\text{pol}}\rangle \tag{4}$$

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# Chemical information from EFG

## A challenging property

Within an atomic shell:

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2. The EFG is sensitive to deviations from spherical symmetry (e.g., chemical bond formation).

# Dailey-Townes model

**C.H. Townes and B. P. Dailey, J. Chem. Phys. 17, 782, (1949)**





The magnitude of $q$ at a nucleus depends almost entirely on the way in which the lowest available $p$-type orbits are occupied by valence electrons surrounding the nucleus. An $s$-type orbit or any closed shell of electrons is spherically symmetric and hence gives no contribution to the variation of energy with nuclear orientation or to $\partial^2 V/\partial z^2$ at the nucleus.

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Tested on: ICl and $UO_2^{2+}$

## Analysis of $\langle e\hat{q}_{\mathbf{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

### 4c-DFT/scalar-relativistic/`dyall.3zp`

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# Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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$$\langle e\hat{q}_{\mathrm{Cl}} \rangle = \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle + eq^{\mathrm{nucl}}_{\mathrm{Cl}} = \sum_{A} \left( \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{princ}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{hyb}} \right) + \sum_{A,B \neq A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A,B)}_{\mathrm{inter}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}} + eq^{\mathrm{nucl}}_{\mathrm{Cl}} \qquad (7)$$

| | | $q$ ($E_h/ea_0^2$) |
|---|---|---|
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{intra}}$ | Cl | |
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| | | | $q$ ($E_h/ea_0^2$) |
|---|---|---|---|
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{intra}}$ | Cl | Principal | 5.890 |
| | | Hybridization | −0.570 |
| | I | Principal | −1.212 |
| | | Hybridization | 0.010 |
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|---|---|---|---|
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| | | Hybridization | −0.570 |
| | I | Principal | −1.212 |
| | | Hybridization | 0.010 |
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl,I})}_{\mathrm{inter}}$ | Cl-I | | −0.017 |
| | | | |
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# Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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| | | | $q$ ($E_h/ea_0^2$) |
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| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl,I})}_{\mathrm{inter}}$ | Cl-I | | −0.017 |
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}}$ | | | −0.619 |
| | | | |
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# Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

## 4c-DFT/scalar-relativistic/`dyall.3zp`

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|---|---|---|---|
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| | | Hybridization | 0.010 |
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl,I})}_{\mathrm{inter}}$ | Cl-I | | −0.017 |
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}}$ | | | −0.619 |
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle$ | | | 3.482 |
| | | | |
| | | | |

# Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

## 4c-DFT/scalar-relativistic/`dyall.3zp`

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| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle$ | | | 3.482 |
| $eq^{\mathrm{nucl}}_{\mathrm{Cl}}$ | | | 1.174 |
| | | | |

# Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

## 4c-DFT/scalar-relativistic/`dyall.3zp`

$$\langle e\hat{q}_{\mathrm{Cl}} \rangle = \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle + eq^{\mathrm{nucl}}_{\mathrm{Cl}} = \sum_A \left( \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{princ}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{hyb}} \right) + \sum_{A,B \neq A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A,B)}_{\mathrm{inter}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}} + eq^{\mathrm{nucl}}_{\mathrm{Cl}} \qquad (7)$$

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| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl,I})}_{\mathrm{inter}}$ | Cl-I | | −0.017 |
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}}$ | | | −0.619 |
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle$ | | | 3.482 |
| $eq^{\mathrm{nucl}}_{\mathrm{Cl}}$ | | | 1.174 |
| $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ | | | 4.737 |

# Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

## 4c-DFT/scalar-relativistic/`dyall.3zp`

$$\langle e\hat{q}_{\mathrm{Cl}} \rangle = \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle + eq^{\mathrm{nucl}}_{\mathrm{Cl}} = \sum_A \left( \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{princ}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{hyb}} \right) + \sum_{A,B \neq A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A,B)}_{\mathrm{inter}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}} + eq^{\mathrm{nucl}}_{\mathrm{Cl}} \qquad (7)$$

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| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl,I})}_{\mathrm{inter}}$ | Cl-I | | −0.017 |
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}}$ | | | −0.619 |
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle$ | | | 3.482 |
| $eq^{\mathrm{nucl}}_{\mathrm{Cl}}$ | | | 1.174 |
| $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ | | | 4.737 |

1. The biggest contribution is $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl})}_{\mathrm{princ}}$.

# Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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1. The biggest contribution is $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl})}_{\mathrm{princ}}$.
2. $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl})}_{\mathrm{hyb}}$ is important.

# Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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2. $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl})}_{\mathrm{hyb}}$ is important.
3. $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{I})}_{\mathrm{princ}} + eq^{\mathrm{nucl}}_{\mathrm{Cl}} \approx 0$

# Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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1. The biggest contribution is $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl})}_{\mathrm{princ}}$.
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4. $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl,I})}_{\mathrm{inter}}$ can be safely neglected.

# Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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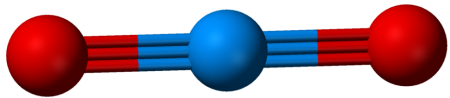
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|---|---|---|---|
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4. $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl,I})}_{\mathrm{inter}}$ can be safely neglected.
5. $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl})}_{\mathrm{pol}}$ is non-negligible.

# Beyond the Dailey-Townes model

## The $[\mathrm{O} \equiv \mathrm{U} \equiv \mathrm{O}]^{2+}$ (uranyl) ion

**U atom in its ground state**
$[\mathrm{Hg}]6p^6 5f^3 6\mathrm{d}^1 7s^2$

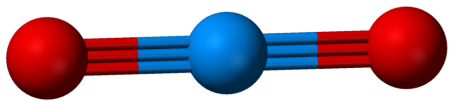


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**Projection analysis picture**
$[\mathrm{Hg}]\ 6p^{5.72}\ 5\mathrm{f}^{2.74} 6\mathrm{d}^{1.05} 7\mathrm{s}^{0.05}$

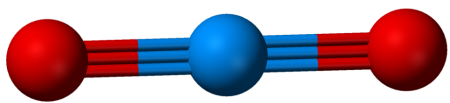


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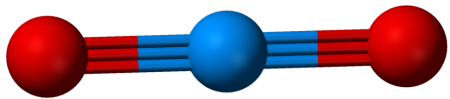
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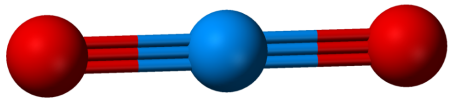
Larsson and Pyykkö, and later Pyykkö and Seth, proposed that the entire $\langle eq_U \rangle$ is provided by U($6p$).

- S. Larsson and P. Pyykkö, *Chem. Phys.* **101**, 355, (1986).
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- S. Larsson and P. Pyykkö, *Chem. Phys.* **101**, 355, (1986).
- P. Pyykkö and M. Seth, *Theor. Chem. Acc.* **96**, 92, (1997).

Subsequent works have shown **additional contributions**.

- W. A. de Jong, L. Visscher and W. C. Nieuwpoort, *J. Mol. Struct.: THEOCHEM* **458**, 41, (1998).
- P. Belanzoni, E. J. Baerends and E. Van Lenthe, *Molecular Physics* **103**, 775, (2005).

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## 4c-DFT/scalar-relativistic/`dyall.v3z`

| | | $q$ ($E_h/ea_0^2$) |
|---|---|---|
| $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle^{(A)}_{\mathrm{intra}}$ | U | -0.579 |
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**Contribution to the EFG arising from the deformation of atoms upon forming the molecule.**

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4. The EFG is highly sensitive to **relativistic** and **correlation** effects.

# Highly Accurate Expectation Values Using High-Order Relativistic Coupled Cluster Theory

G. Fabbro, J. Brandejs and T. Saue, *J. Phys. Chem. A* **129**, 6942, (2025)

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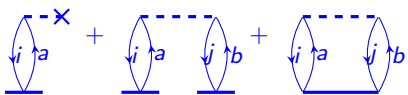
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CC energy with diagrams: **easy!**

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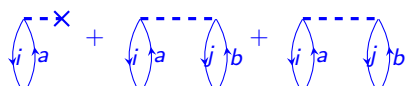
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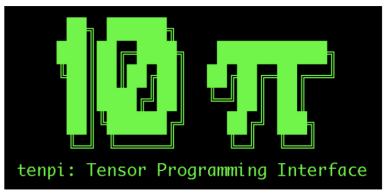
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CC amplitude with diagrams: **often tedious**

1. High-order CC equations involve many terms.
2. Difficult to properly define intermediates.
3. Prone to errors.

# tenpi: elegant development of CC methods

J. Brandejs, J. Pototschnig, T. Saue, J. Chem. Theory Comput., 21, 7320, (2025).




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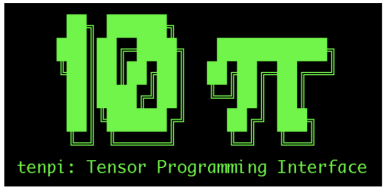
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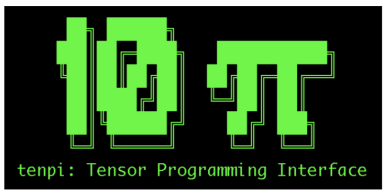
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## Functionalities

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Properties are computed using the Lagrangian framework

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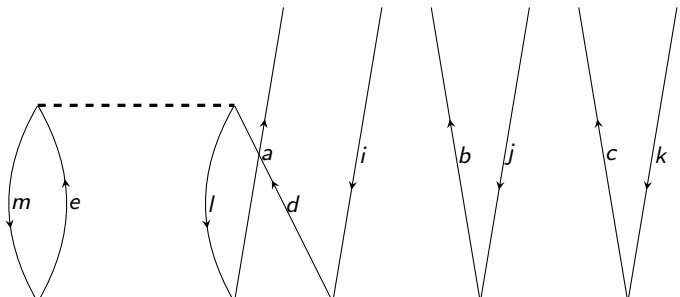
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First-order properties

$$\left.\frac{dL^{\rm CC}}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left.\frac{\partial L^{\rm CC}}{\partial \varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \sum_{pq} h_{pq;A}\ \gamma_{pq}^{\rm CC}\ , \quad \gamma_{pq}^{\rm CC} = \langle\Phi_0|(1+\hat{\Lambda})e^{-\hat{T}}\{a_p^\dagger a_q\}e^{\hat{T}}|\Phi_0\rangle \tag{14}$$

# Diagrams for CC first-order properties

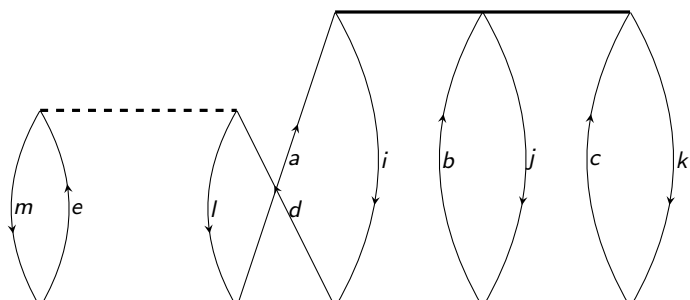
- From $\langle \Phi_{ijk}^{abc} | \hat{\bar{H}} | \Phi_0 \rangle = 0$ we take, as an example, the following diagram



## Diagrams for CC expectation values

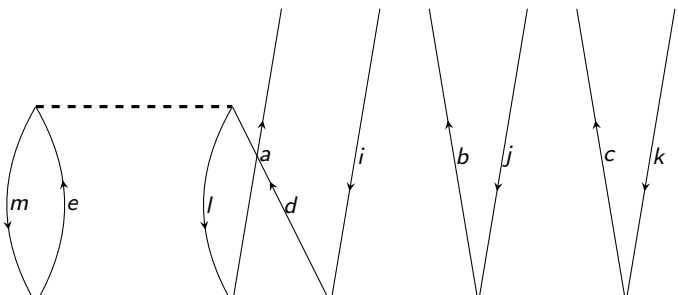
- We introduce the CC Lagrangian:

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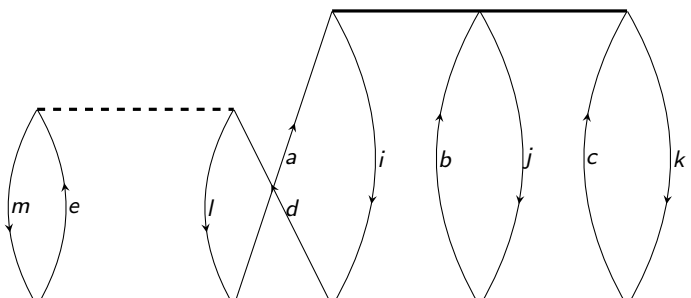
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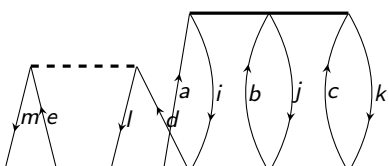
- CC Lagrangian

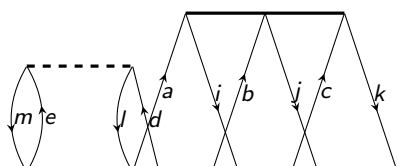
$$L^{\mathrm{CC}} = E^{\mathrm{CC}} + \sum_{\mu} \bar{t}_{\mu} \langle \Phi_{\mu} | \hat{\bar{H}} | \Phi_0 \rangle = \langle \Phi_0 | (1 + \hat{\Lambda}) \hat{\bar{H}} | \Phi_0 \rangle \,, \qquad \hat{\Lambda} = \sum_{\mu} \bar{t}_{\mu} \hat{\tau}_{\mu}^{\dagger}$$



# Diagrams for CC first-order properties

- $\Lambda-$equations: $\frac{\partial L^{CC}}{\partial t_\mu} = \langle \Phi_0 | (1 + \hat{\Lambda}) [\hat{\bar{H}}, \hat{\tau}_\mu] | \Phi_0 \rangle = 0.$



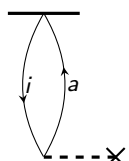



## Diagrams for CC first-order properties

- $\langle \hat{h}_{\mathrm{A}} \rangle = \sum_{pq} h_{pq,A} \underbrace{\langle \Phi_0 | (1 + \hat{\Lambda}) e^{-\hat{T}} \{ a_p^\dagger a_q \} e^{\hat{T}} | \Phi_0 \rangle}_{\gamma_{pq}^{\mathrm{CC}}}$

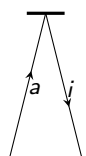
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- $\langle \hat{h}_A \rangle = \sum_{pq} h_{pq,A} \underbrace{\langle \Phi_0 | (1 + \hat{\Lambda}) e^{-\hat{T}} \{ a_p^\dagger a_q \} e^{\hat{T}} | \Phi_0 \rangle}_{\gamma_{pq}^{CC}}$
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- Substitution: $\hat{h}_\mathrm{A} \rightarrow \hat{F}_N$
- $\gamma_{ai}^{\mathrm{CC}} = \frac{\partial \langle \hat{F}_\mathrm{N} \rangle}{\partial f_{ai}}$



# Correctness: dipole moment of LiH

| Method | Code | Convergence | Total dipole moment [$ea_0$] |
|---|---|---|---|
| SCF | DIRAC | | -2.3326823**2** |
| SCF | MRCC[1] | | -2.3326823**3** |
| CCSD | DIRAC `tenpi` | 0.6E-09 | -2.1655380**0** |
| CCSD | MRCC | 1.3E-10 | -2.1655380**2** |
| CCSDT | DIRAC `tenpi` | 0.9E-09 | -2.1652806**4** |
| CCSDT | MRCC | 1.8E-10 | -2.1652806**6** |
| CCSDTQ | DIRAC `tenpi` | 0.6E-09 | -2.1652794**3** |
| CCSDTQ | MRCC | 2.1E-10 | -2.1652794**5** |

Table: Convergence of the final error shown in the run and total dipole moment for different methods and codes for LiH.

[1] Kállay, M. et al., *J. Chem. Phys.* **152**, 074107 (2020)

# Applications: NQMs of $^{7}$Li and $^{27}$Al from molecular calculations

## Computational protocol

- The exact-two-component molecular mean field (X2Cmmf) Hamiltonian has been used.
  - J. Sikkema, L. Visscher, T. Saue, M. Iliaš, *J. Chem. Phys.* **131**, 124116 (2009)

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where `aug-basis` is `19s11p6d3f` for Li and `25s16p11d8f5g` for Al.

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- Finally, we add the **rovibrational corrections**

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- Extraction of $Q(^{7}\text{Li})$ and $Q(^{27}\text{Al})$ by using the **orthogonal distance regression (ODR)**.

# Applications: NQMs of $^{7}\mathrm{Li}$ and $^{27}\mathrm{Al}$ from molecular calculations

## Results

$$\boxed{Q(^{27}\mathrm{Al}) = 0.146598 \pm 0.000001 \text{ b}}$$

- $\tilde{T}$ contributes with $\approx 0.75\%$.
- $\tilde{Q}$ is negligible.
- $q_{\text{rovib}} \approx 0.8 - 0.9\%$.

Perfect agreement with the current accepted value
$(Q(^{27}\mathrm{Al}) = 0.146(4)$ b).

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However, in perfect agreement with the results reported by Guan et al. ($Q(^{7}\mathrm{Li}) = -0.0386(5)$ b).

- H. Guan et al., Phys. Rev. A 112, L010801 (2025).

# From Single-Node to Multi-Node

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**Nowadays: Multi-node extension using TAPP**

DIRAC (Fortran) $\xleftrightarrow{\text{TAPP interface}}$ Cyclops CTF (C++)

- **Cyclops CTF**: is a distributed memory tensor contraction library designed for HPC.
- **TAPP**: Tensor Algebra Processing Primitives;
  allows quick integration of tensor libraries into application codes.

# Revised value for the nuclear quadrupole moments of $^{87}$Sr from relativistic molecular calculations

G. Fabbro, J. Brandejs and T. Saue, In preparation.

# $Q(^{87}\mathrm{Sr})$ from SrO and SrS

## First molecular attempt

- **Sahoo et al.**: $Q(^{87}\mathrm{Sr}) = 0.305(2)$ b.
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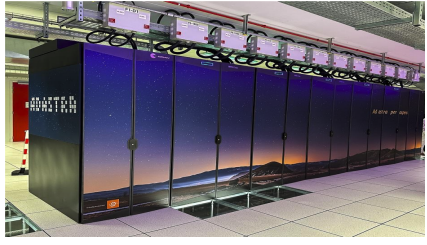
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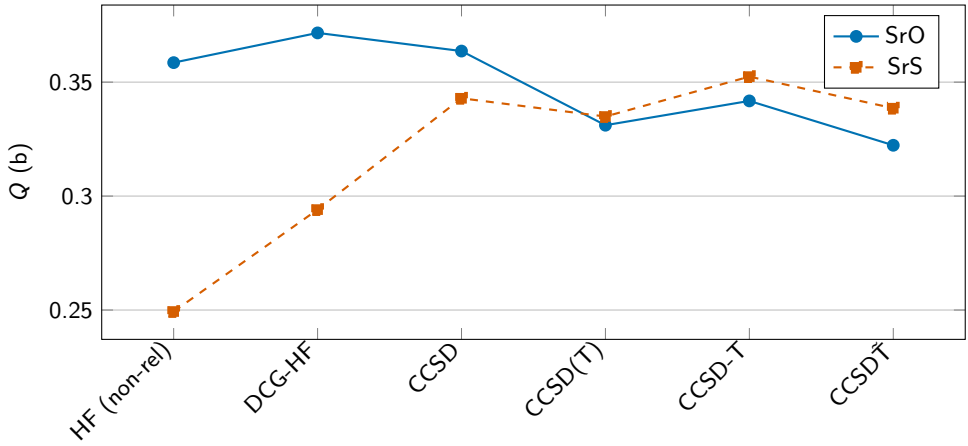
CCSDT calculations were performed on the **Genoa partition of the French Adastra supercomputer**.

We correlated the highest eight occupied spinors and the lowest **116 virtual spinors**.

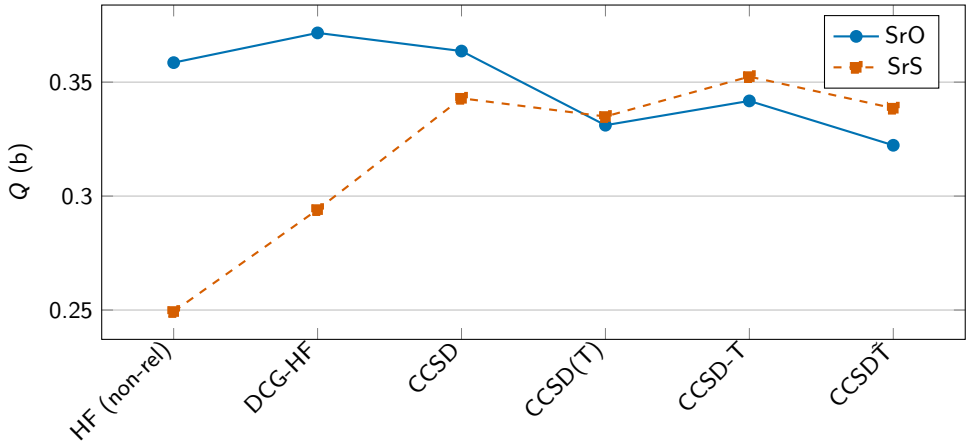
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# $Q(^{87}\mathrm{Sr})$ from SrO and SrS



# $Q(^{87}\mathrm{Sr})$ from SrO and SrS




We found that the CCSD(T) provide the most balanced description across the SrX series

$$Q(^{87}\mathrm{Sr}) = 0.33493 \pm 0.00165 \text{ b}$$

# Conclusions

**Method Development**

1. The **tenpi toolchain** has proven effective for generating $t/\Lambda$ equations for arbitrary CC level.
   - First implementation of CCSDT/CCSDTQ expectation values in the DIRAC code.
2. **Multi-node extension** of high-order CC codes using Cyclops.
   - Realistic calculations for heavy atoms with large basis sets and active spaces.

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### Applications

1. The **Dailey-Townes model** alone is too crude to provide a reliable understanding of the EFG.
   - It is a particular approximation to our projection analysis.
   - Hybridization, relativistic and core polarization effects must be accounted for.
2. High-order CC expectation values enabled accurate EFGs, yielding **updated NQMs**.

# Perspectives

- Inclusion of relaxation effects

$$L^{\mathrm{CC}} = \langle \Phi_0 | (1 + \hat{\Lambda}) \hat{\bar{H}} | \Phi_0 \rangle + 2 \sum_{ai} D_{ai} f_{ai} + \sum_{pq} I_{pq} \Big( \sum_{\mu\nu} C_{\mu p} S_{\mu\nu} C_{\nu q} - \delta_{pq} \Big)$$

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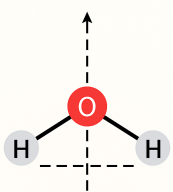
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Point group: $C_{2v}$

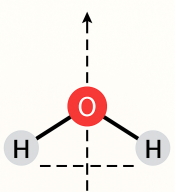
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- High-order molecular properties.

- Implementation of new methodologies using tenpi:
  - EOM-CC;
  - TD-CC;
  - UCC;
  - ...

## Acknowledgements

With thanks to

- Trond Saue
- Jan Brandejs
- HAMP-vQED team

**and all of you for the attention!**









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